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E-COMMERCE AND CYBER LAWS

As e-commerce solutions, its application and business models converge in new ways, atidal wave of change is transforming industry, redefining competitive strategies that serve, delight and retain customer better than everbefore.

Impact of Change

A modern business which addresses the needs of organizations, merchants and consumers to cut costs while improving the quality of goods and services and increasing the speed of service delivery and use of computer networks to search and retrieve information in support of decision making. The science of computing and information systems have emerged as the major change in business and in society. There is a revolution taking place in electronic commerce. Global networking and other broadband technologies are being used as competitive weapons in today's business. Under pressure from technological change, competitive and decreasing prices of hardware, the world of network infrastructure began a speedy transformation to New technological advancement, local area networks, robust, protocols, routers and user friendly software found market place that motivated individual initiative to experiment with networking. Customers today expect for greater degree of change in products, brands and services offered than has been the case in the past, we live and work in a period of continuous business transformation through the application of technology and marketing challenges as fashion and product life cycles become shorter. Change has become the norm. The environment has seen profound changes as a result of factors—changing customer attitudes, increasing competition, new technologies and information overload. We must take advantage of every opportunity to improve our knowledge and skills. Opportunities exist to use collective strength to get the best from challenging markets. To disseminate information on new development and changing requirements on a global scale to enjoy success and prosperity in then ear future.

Electronic Commerce has become critical due to:

(a) Customer to business interactions,

(b) Intra business interactions, and

(c) Business to business interaction.

Electronic Commerce is enabling the Customer to have an increasing say in what products are made, how products are made and how services are delivered to a faster and more open process with customers. Electronic Commerce is a catalyst change in internal organizational function as evidenced by the rapid proliferation of Intranets. We witness a shift from a hierarchical command and control organization to the information based organization. Electronic Commerce facilitates the network form of organization when firm rely on product distribution to meet customer demand more management solution is a desirable goal that is needed to manage the chain of network linking customers, workers, suppliers, distributors and even competitors. Modern managers can operate effectively with some rearranging of mind sets, skills, knowledge and wisdom.

Objectives

Electronic Commerce is a new way towards conducting, managing and executing business transactions using computer and telecommunication network. Electronic commerce is expected to improve productivity and competitiveness of participating business by providing access to an online global market place with millions of customers and thousand of products and services. Another objective is to provide participating companies with a little cost and time with efficient means for working with customers and suppliers and development partners. If every one works as planned network based e-commerce, this would enable companies to shorten procurement cycles, cut costs and shrink product development cycles and accelerate time to market through engineering and product customization. The electronic commerce market place is expected to support all business services that depend on paper based transactions. The use of a network based infrastructure reduces the cost and levels for both small and large applications. This consists of electronic commerce applications including both inter and intra organizational and electronic market. The building blocks of the infrastructure such as a common business information services, messaging and information distribution and the information super-high. The public and technical standards would be necessary to support the applications and the infrastructure.

New Pressures

The reaction of business pressures are such that companies are restructuring. There is a great need for faster reaction time to environmental events, customer requirements, competitive new products and services in order to achieve excellent results. E-commerce is being established in the areas of business world and the results would be imminent and the changes not only in the society and the economy at large. The information society presents Governments with an urgent need to consult with broadest range of

participants and particularly with corporate users and consumers. Information and Information Technology are the new drivers of this age. Technology is an enabler of change has to be driven by business executives with a view to take advantage of technology and environment.

Microsofts New Experiences

Microsoft anticipates changing customer experiences by re-engineering several value chains including travel, automotive sales, real estate and finance. The success of the new segregated value chain depends on an integrated infrastructure that yields easy to use and sophisticated customer interaction. The target of these services are vast. Microsoft plans to win a major share of advertising market and also sales and distribution in the market, for airline tickets, automobile sales and retail goods. Thus Microsoft is creating an entirely new set of service dynamics in various industries in turbulent weather.

Beauty of the Internet

The Internet is about the sharing and exchange of ideas in a free manner than about data transfer between machines. The Internet has transformed the way network infrastructures were built around highly structured architectures that were planned, created and operated by Government, Telcos and big corporations. The basic infrastructure flowed top down from formal meetings and standards that purported to chart the future requirements. Commercialization and privatization of the Internet through commerce service provider have led to the development of electronic Commerce. The ability to do business over the Internet benefit the Internet community, fosters market efficiency by facilitating small vendors entry into markets by providing consumers with broader range of choices. The Net is an interconnection of computer communication networks, spanning the entire global, crossing all geographical boundaries and touching life styles and so on.

The Network Infrastructure

An infrastructure for conducting e-business would enable automated transactions between consumers and service providers ranging from department store, mail order store, broad cost networks and even business to business commerce. This is due to the development of new telecommunications technologies and the continue development of fiber optic facilities which have resulted in higher transmission speeds at significant lower costs. The end result is a seamless web called the I way of communication network, computers, digital libraries and consumer electronics that put vast amount of information at user's finger tips. An open competitive architecture is the desirable long-term objective but there is a trade off between open network and incentives for large investments. An open environment is good for consumers because it promotes the introduction of new services, its application and promotion at lower prices. An open environment is good

for the industry because it promotes innovation and cost reduction which in turn leads to competitiveness. Since the benefits of this technology flows primarily to consumers of I way application and services and typically do not produce a propriety advantage for any I way application and service provider.

E-commerce and the World Wide Web

The world wide web companies have established store windows to show off their wares. WWW server have created a flourishing commercial district in cyberspace. The WWW provides a standard set of protocols for presenting and retrieving information. It allows users to jump automatically from one data base to another via hyper links without having to know where information is located as large well known companies have a presence on the WWW.

Integration

Electronic Commerce and the world wide web leads to the integration of network infrastructure, data management and security services to allow business applications in different organizations to interchange information automatically. Communication services transfer the information from the originator to the receipt. Data management services lead to the interchange format of the information. Security services authenticate the source of information, verify the integrity of the information received by the receipt, prevent disclosure of the information to unauthorized users and verify that the intended receipt received. The world wide web is an ideal mechanism for providing relevant information to the public globally. It allows organizations an opportunity for publicizing their products and services at minimal cost. By putting information on routinely asked questions on the home pages, organizations are saving costs by reducing the number of customer service representatives. Also customers place orders through the Internet once security issues are worked out. The WWW holds the potential to increase the market share and helps expand into new markets by virtue of its global reach. Competition is much more fierce than ever before, to increase productivity, efficiency, reduce the costs of products and services and the use of information technology to innovate in order to expand and create new markets and retain customers better than before.

The web is used towards attracting new customers. Strengthening existing customers, developing new markets and distribution, developing new information based products. The Web architecture facilitates electronic commerce by providing a set of tools which can make interactive applications come to life.

Browser	Network	Server
Click on any object	Request	Stored object
Cannot read format	Request	Stored Applet
Load applet	Reply	Send payment in applet use

Supply Chain Management

E-commerce and online business, an entire new industry called the Internet service Provider has emerged which provide a door through which users travel to reach the content. The steady commercialization and privatization of the Internet through Commerce Service Provider fortells the growth of electronic commerce. The ability to do business over the Internet would benefit the Internet community as it would foster market efficiency by providing customers with broader range of choice. Clearly companies and consumers are discovering that global networking and other technological innovations are powerful assets used as competitive weapons in day to day business activities.

Supply Chain management refers to the integration of the internal and external partners on the supply and process chains to get order from customers, raw materials from suppliers and offer finished products to end users. SCM enables companies to avoid the vagaries of the market place by improved stock management techniques and better distribution management. Effective management of inter firm linkages in the supply chain add value in numerous ways, lower cost, improved quality, faster speed of delivery, improved support and stronger competitive positioning.

Intranet

Intranet based electronic commerce technology and supply chain functions are interwired as the wall between functional structures are dismantled. Software development that provided isolated solutions to the problems of the various divisions is a thing of the past. The focus today is on the optimization of integrated processes that extend beyond particular divisions, taking account of the customers wishes. Intranet technology that enables the exchange and linking of data from all areas of the company. The Intranet environment offers a new approach to better address certain business requirements. Electronic commerce application are being built on a foundation of global hypertext developer that hypertext is just another user interface approach. The web is a net working technology plus an information management method that provides a novel way of directly accessing and manipulating information. It is an interface that integrates security and payment schemes which can be used for on line buying and selling. Thus the web provides a totally different and unique method of accessing information. Where as traditional data bases have some structure to them, a hypertext data base has of regular structure. The user is free to explore and assimilate information in different ways, an immense flexibility that is making electronic commerce a reality.

Electronic Payment Technology

Electronic payment system are widely used in commerce including whole sale payments, wire transfer, recurring bill payments, the automated clearing house, electronic draft capture and electronic check payment. The emerging area of cash less payment and the

various payment options that are currently being available. These payment options include e-cash, electronic checks, smart cards and electronic purses. A world with electronic payment system would be different from today's world. New opportunities would arise for consumers, banks and others. If record keeping were part of the payment instrument, several innovative uses could be visualized. Customers could tally up every thing they spent with a little effort. They could do realistic expenses analysis and even negotiate with merchants based on volume. Even small merchants could know who utilizes their stores and how much they spend. Possible benefits to banks and other firms would be a better understanding of the financial behaviour of their customers. Governments might monitor prices, inflation and velocity of money more accurately and more quickly than today. The challenge for the future would be to come to the right vision how electronic payment systems would augment or replace paper based payment methods. The next step would be to know exactly how to make a profit on this account. EC Applications is to foster an environment in which public confidence in security arrangement for electronic transaction would grow. It is a matter of employing the best possible available security devices in networks that are designed with security in mind. It is also matter of providing an institutional framework that can support.

Authentication/Data Integrity
Certification and Data Protection

Security Plans, authentication techniques, access controls, confidentiality measures and data transmission controls are all considerations for electronic payment systems. An understanding of the key components, underlying risks, key control features and workable procedures to be obtained about safeguarding assets and payments in a cost effective and secure way.

The Net is changing everything from the way conduct commerce, to the way we distribute information. Being an interactive two way medium, the net through innumerable web sites enables participation by individuals in business to business and business to consumer commerce, visit to shopping malls, entertainment sites and so on in cyberspace. The Internet now enables new ways of delivering Government services to its citizens in their homes and offices at various levels, interacting with for filling documents or receiving information from and are undergoing changes through the technology of the web and the Internet use of Electronic Mail (e-mail) for inter-personal messaging, the need to use information and telecommunication by organizations. Internet Service providers and the National IT Task Force set up towards building blocks for EC implementation. Networked global society where barriers and tariffs are being dismantled, nations are building information infrastructures.

Network Security

There is an increasing demand for efficient collection, dissemination and processing of information due to various economic and market forces. Electronic commerce makes

use of today's technology to solve business communication and Computer intensive business problems. Today's computer systems are linked global networks which face a variety of security threats that result in significant information and financial losses. Threats vary considerably from threat to system availability from malicious hackers attempting to crash a system to data or transaction integrity threats. An understanding of the types of threats in today's computing environment can assist a security manager in selecting appropriate cost effective controls to protect valuable information resources. An overview of many of today's common threats would be useful to organizations studying their own threat environments with a view to develop proper solutions specific to their organizations.

Security is always an issue the database contains proprietary information. We might transmit technological information that our competitors would want so we need to make sure that data is secure. Security as an issue of concern that it is possible to overcome by using techniques such as Encryption and Firewalls. Encryption is a reversible process of modifying clear text for the purpose of the keeping it secret from anyone other than its intended recipient. The firewall may include one of more gate ways or routers and may comprise separate network segments as well as software filtering and other mechanisms for protecting corporate network resources.

Security Requirements of EDI Messages Involves

EDI message integrity so that any change of content can be detected. EdI message origin authentication so that the recipient can verify that the contents of the EDI—Message are identical to that sent by the originator.

EdI Message confidentiality so that no unauthorized entity can read the actual content. Security service can be provided by the transport system *i.e.* by the network service provider. A security policy would have to be developed by organizations so that the effective security can be established. Nations need to work out their position in the form of policy statements and while paper on cryptography and encryption for promoting electronic commerce.

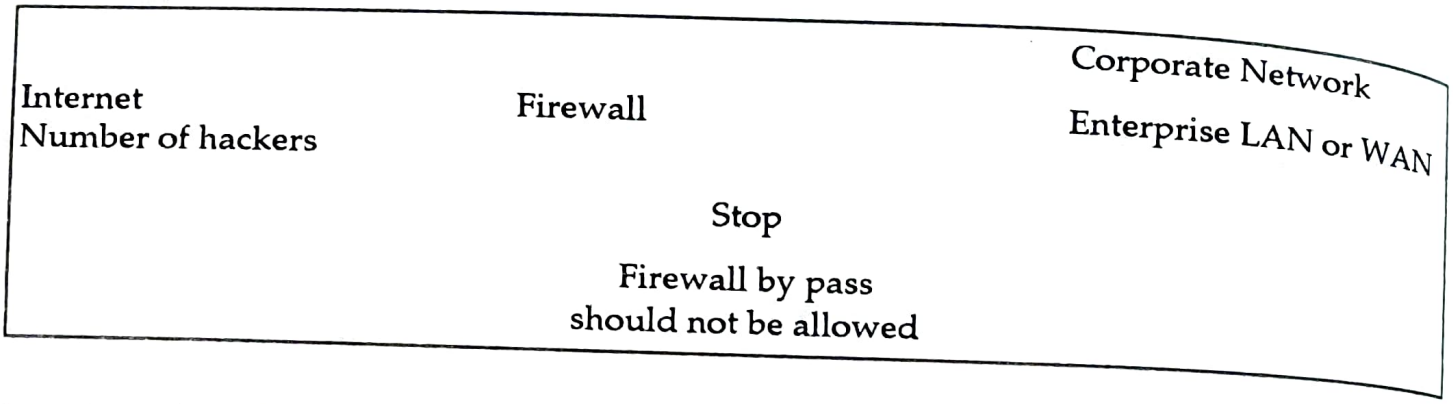
Firewalls are built to protect the internal network of an organization from attach originating from the Internet. Firewalls can be implemented by:

(a) Establishing rules to decide which packets depending on the originating IP address should be allowed to pass.

(b) Establishment of an additional network as a buffer between the internal and external network.

Establishment of proxy servers so that internal clients for accessing external services are routed through the proxy server.

Firewall Secured Internet Connection



Legal Implications

The role of the Government is to provide a legal framework for electronic commerce so that while domestic and international trade are allowed to expand their horizons, basic rights such as privacy, intellectual property, prevention of fraud and consumer protection etc. are taken care of. The free flow of information through the use of Internet without nay geographic and national barriers comes forth with a myriad security and privacy problems. There are legal issues with respect to patents, trade marks, copyrights, responsibility of ownership of data and contracts and so on. This calls for changes in the legal economic and commercial framework. Electronic governance uses the Internet to deliver services to citizens. Information can be retrieved, forms can be submitted, returns can be filled electronic, Government is transacting business with the public in areas as taxation, issue of documents, passport and driving licenses etc.

New Challenges

E-commerce is being established in all areas of the business world. The information infrastructure responsive to the needs of commercial and non-commercial users, there is a need for substantive actions. We urge Govt. to formulate policy in consultation with users and to pursue these initiatives to address legal issues and to provide means of redress for those suffering from undesirable acts penerated in the environment. The removal of administrative and regulatory barriers to technological convergence and the provisions of an open and easily accessible networking environment are the key issues of competitiveness. Both industry and Govt. need flexibility in order to respond productivity and rapidly to the new challenges of E-commerce. Commercial development including encryption software to safeguard data transmitted over the Internet and programmes and the transformation of the Internet into a primary route for electronic commerce.

E-Commerce in India

With the rapid expansion of the Internet, Global Electronic Commerce is set to play an important role in the twenty first century in India. The new opportunities that are open

are accessible to both large corporations and small companies. Consumers would be able to choose from a mind boggling array of products and services. However, the right commercial and legal environment is crucial to this material barriers. Consumers and merchants interests have to be protected. The role of the Government is to provide a legal framework for electronic commerce are allowed to expand their horizons and basic rights are taken care of it.

EDI in Indian Customs

The Indian Customs EDI Systems (ICES) designed, developed and implemented jointly by the National Informatics Centre and the trade in the Country. ICES has transformed the custom house into a paperless office. Electronic submission of documents. The CHA's can use the Remote EDI System (RES) which is a standard one software package for preparation of Bills of Entry (BES) and shipping bills and other related documents. It runs on PCs equipped with a modem and a telephone connection for connectivity to NICNET. The trading community is happy about the transparency of procedures and new processes and the accountability that goes with it. The benefit of ICES are as follows:

Transparency of the system
Faster clearance of consignment
Online Enquiry
Uniform assessment
Electronic submission of documents
Better efficiency in customers operations

ICES provides for integration of other agencies involved in the customs clearance through EDI technology over NICNET.

Indian Companies

Indian companies like Hindustan Lever Ltd., AVB Group, ITC, TISCO, Eicher and Reliance in the private sector and Bharat Heavy Electrical Ltd., C DOT, Indian Oil Corporation etc. in the public sector etc. are using information technology in view of global competition in order to gain the competitive advantage. New concepts like telemarketing, advertising on the Internet, teleconferencing and telecommunication etc. are the new ways of conducting modern business Kumar Mangalam Birla is not only preserving the business Kingdom handover over by Dr. G.D. Birla, Aditya Birla and AVB Group constitutes the multi product diversified companies and Kumar Mangalam's belief in "Institutionalizing procedures and System Orientation". The logo has been changed to Rising Sun in order to impart a feeling of identification for group companies.

The focus on e-commerce and e-business is to stay in the private sector and public sector companies and continuously upgrading and improving total quality and quick delivery of service while reducing overall cost in existing business. Thus available resources are within the group/groups have been ploughed back an investments into

the core group business. The emerging team paradigm is constituted on the assumption that knowledge and wisdom found among the members of the organization when brought together as team of experts. Thus designing systems that are competitive and efficient and creating an information architecture that supports the organizational goals, and use the same in the socially and ethical responsible manner as well as use of information technology for growth and development.

Reliance has been maintaining its status by mobilizing resources for innovation in design, plant modernization and sales promotion. This generates enough cash for the company. Exports provide the much needed foreign exchange in order to update its technology through latest machinery to maintain its Technological superiority. Reliance's unique marketing strategy and its successful implementation includes market segmentation, matching marketing needs with product features, advertising and sales promotion and commitment of dealers. Reliance adopted innovative financial strategy. The company mobilized funds from capital markets through issue of convertible debentures. Reliance being a zero tax company in spite of high profitability and have taken massive expansion programme. This provides golden opportunity to the company to achieve its mission and goal even in a highly competitive global market. On account of this, new opportunities have arisen which are being exploited in a profitable manner what their customers dealers, employees etc. anticipate in the near future.

Arvind Mills, Ahmedabad—New Vision

New Business visions of Arvind Mills, Ahmedabad is a market oriented organization depicts new vision, building quality into process, empowerment and teamwork. Arvind Mills have information about Customer needs, product and competitions, global prospects through the use of information technology that serve customers, employees and management in the new information era. This new vision includes:

- Building positive attitudes and values towards information sharing and it appropriate uses.
- Valuing employees with commitment and use of knowledge in action, commitment to Quality, trust in people as well as development of organization in a continuous way.
- Moving from brain drain to brain gain.

New Challenges towards new discoveries into best weapons of humanity to fight the poverty and unemployment in the 21st century.

Reliance's Mega Dream into Reality

The convergence of the information and communication business as the Infocom Revolution. Reliance's new vision to be leading investor in infocom initiatives. The all optic fibre network would provide value added voice, image, data services, the broad

based network, internet data centers would provide e-commerce media casting web hosting and software services.

This includes Reliance wants to be king of power and inforcom sectors wants to be global player for tie-up in the entire digital value chain. Internet net mesh linking 115 cities with terabit band. Every service to be available through all optic fibre network.

TCS Strategy

TCS Strategy is in terms of creating and owning knowledge is of course one strategy. The other is to license the knowledge and capability with some body else and thirdly to buy the intellectual assets and to inculcate technologies and ideas of people both within TCS and outside. Regarding owning knowledge, the key is to build on it and know-how to market it globally. TCS imparts training people on multiple technologies and assimilate new technologies within the organization. There is always appreciation for the need for systematic and continuous improvement in Quality—learning to win in terms of companies to win the market place and the customer by systematically building skills. There is a technique involved in the process, response time involved in the speed, determination in the willingness to succeed and customer sensitivity involved in flexibility, strategies to compete and to, plan carefully for implementation, communicate the strategies and plans for new vision. Community initiatives and social responsibility is a major dimension. The opportunity to learn new technologies and accreditation as Quality analysts and their opportunity located in any part of the world. Commitment, drive, leadership, vision and setting a personal example of integrity have been the ingredients of success of an organization. TISCO has already let in its HR programme and employee training in quality to become globally competitive.

Challenges & Opportunities E-Commerce

India entered the age of E-Commerce when the Govt. declared the Internet Services provider (ISP) policy allowing private players into the market. Although the Indian Figures are by means compatible to the Amercian ones, we continue to show a good growth. For information, Madura Garments, Charagh Din, India Faship, Maberix Information services and the RPG Group have scrabled to the among the first to set up net shops, HLL, Titan, Times, Godrej, Trigger, Jeans, Shopper's stop and Nilgiri are waiting for the base to widen before they look at the net. Nilgiri is an up market department store chain in Bangalore starting with birthday cakes and dairy products offer over the net. Mumbai, Bangalore and Hyderabad is also experiencing a similar story. Attempts to sell vegetable/fruits on the net are going to work in the long run. Nilgiri's is the a fact as little as Rs. 50,000/- which is small price for attracting customers. Equally impressed is Mumbai, Panwallah who has more then 18,000 visitors to his website. Mumbai used Annet Communication coupled with media interest has the same bookstore proprietor's banging on Bulchandani's door.

In India, despite not having a telecom or cable infrastructure, however, the total value of the India E-Commerce transactions during 1998-99 was Rs. 131 Crore (Rs. 119 crore in business to business, or Rs. 12 crore). If business see E-Commerce as a liable proposition, its massive potential could be unleased if we have a supportive framework in place. NASSCOM estimate that if certain changes take place, the value of e-commerce transactions in India which is very much in sync with the global pattern.

To talk about a bright future for e-commerce in India, perhaps, major stumbling block would be the adequate telecom infrastructure outside the metros as well as the cost and difficulty of getting leased lines and bandwidth for an ISP. Now existence of cyber law pertaining to E-Commerce transaction is another major factor. PC penetration is an issue as PC, continue to become cheaper. The Govt. could do to promote e-commerce in India would be to make its transaction is another major factor. PC penetration is an issue as PC continue to become cheaper. The Govt. could do to promote e-commerce in India would be to make its transaction sales tax free for five years. There is another dimension to the security of data during transit that is at stake. Responding to such fears world wide, alternative payment modes have emerged. One is the Secure Electronic Transaction (SET) developed by Global Credit Card Franchisers via and Master Card. City Bank is working with another data encryptions.

The Information Technology in every day life is in the public eye now more than ever before because of a number of issues in which technology and national policies are closely related.

- As the use of informational technology expands, there is a continuing need for e-commerce and e-business which can be aligned to achieve business goals and support business imperatives. (There is an increasing demand for efficient collection and processing of information due to various economic and market forces). Application of electronic commerce techniques and e-business designs and strategies to solve business communications and Computer intensive business problems.

Examples relate to the current international business environment, management practices and programmes offer progressive managers with new options for lowering costs, restructuring work flaws, streamlining operations and redefining strategic focus.

Companies are discovering that old solutions donot work with new problems as the business parameters have changed. Electronic Commerce has become critical with new dimensions—Customer to business parameters have changed. Electronic Commerce has become critical with new dimensions—customer to business interactions, intra-business interactions and business to business interactions. In order to be successful, management has to come to grips with the changes taking place and managers can operate effectively with re-gauging of mindset, attitudes, skills and knowledge in shaping the future of e-commerce and e-business.

The Internet and The Access Provider Industry

The Internet and its commercialization and the service providers industry are responsible for the explosive growth in Internet connections. Economic and social trends driving the need for data access and exchange that are gaining momentum daily, opening up exciting new opportunities for access providers worldwide. However, a great deal of uncertainty spans a variety of management issues including marketing and pricing, consumer behaviour and technology. Market uncertainty creates the potential for shaping new products and markets and building a loyal customer base. The ways in which access providers would exploit and shape this immature market to be seen in the near future.

Web Technology

As web technology and practices mature, integration if corporate data bases and applications is sure to play an expanding role and the web as important part of the corporate infrastructure to meet new challenges about coordination and efficiency needs without destroying the independence of decision making and action that makes enterprise flexible. By allowing data to be exchanged freely among everyone linked to the corporate web, intranets solve a basic problem for information systems. Prior to the web, almost every time a software application on a corporate network was added on changed through a lengthy deployment process of reconfiguring the network hardware and software. Today an application deployment as the user interface (the web browser does not change, the end user training is minimal and systems managers donot have to worry about content or network in compatibilities. Web sites are easy to build and grow quickly, information systems put into place policies to incorporate them into the corporate information infrastructure and to manage the PCs and local area networks. Organizations need scalable solutions that keep pace with the increase in web sites. The role of IS management is changing from controlling gatekeeper to enabling gateway and some form of data and document is needed to facilitate this information. In addition to tools such as search engines, administrative tools are required to help the systems managers of the internal web sites. To identify how the web affects the business and learning how to use this new technology in specific business processes and management is always concerned about costs, productivity and profitability. The market for web software and services has begun to develop and is characterized by increasing number of market entrants. A new and rapidly evolving industry, demand and market acceptance for new products, demand and market acceptance for new products and services are subject to a high level of uncertainty. Clearly the opportunity for companies developing web software is great and the challenges are quite difficult.

What exactly is Java?

Java is a simple, object-oriented, multi threaded garbage—collected, secure, robust, architecture neutral, portable, high performance, dynamic language. The language is

similar to C & C++ but much simpler. Java programmes are compiled into a binary format that can be much simpler. Java programmes are compiled into a binary format that can be much simpler. Java programmes are compiled into a binary format that can be executed on many platforms without recompilation. This binary format is called byte code independent. The language contains mechanisms to verify and execute Java programmes in a controlled environment, protecting the user's PC from viruses and other security violations.

Role of Java in Electronic Commerce

Java adds an element of intelligence and computation paver to existing web browsers are relatively two dimensional viewers, they offer little in the way of interactivity or computing power. With Java compatibility, a browser becomes a computer capable of executing software programmes and communicating with other Computers. More important Java enables content to become active. Today web browsers allow users to manipulate information spread across the Internet as a whole. Web browsers integrate the function of fetching the remote information, figuring one what format it is and displaying it. These browsers contain detailed hard used knowledge about the many different data bytes, protocols and behaviours necessary to navigate the web changes require that the user delete the old browser and reinstall a new version.

A view of Integrated Browsers

HTTP FTP GOPHER NNTP HTML IPEG

A view of Integrated browsers—captures the problem associated with hard coding functionality, if a user wants to access data on multiple server search having a proprietor, HTML extension as specialized application are being implemented using propriety.

The dynamic incorporation of applets has significance to have business is done in electronic commerce environments. With applet based browsers as a base, vendors can produce and sell exactly the price that is added value to what exists and integrate it smoothly with the products of other vendors. This seamless integration creates a final result that is convenient for the end user a necessary condition of electronic commerce.

Business Reasons for using Java

There are clear technical, cost and marketing advantages to develop applications in Java:

- Ease of destination
- Speed of development
- Developer critical mass

- Fits the web paradigm and
- Third party support

Independent developers view the emergence of Java as an opportunity to depart from the Wintel platform hegemony. Java is a tool that has the potential to dislodge some of the dominant, platform or operating system. Centric software companies like Microsoft. In essence, Java has leveled the playing field for many developers both large and small.

Internet and Web-based Telephony

Telephony has become the fastest growing type of service on the Internet. Microsoft has planned to embed Internet voice technology into its Windows Operating Systems and Explorer software. Intel has added multimedia extensions to its future PC micro processors. Compaq Computer and other leading PC Makers are building voice capabilities into their machines. The proliferation of web sites presents users with vast amount of information. With the amount of information increasingly rapidly, there is a need for tools that assist in searching for relevant information. Among the better known search engines that provide customers with time saving and cost cutting services are Yahoo, Lycos and AltaVista. Lycos is a search engine based on Lycos Spider Technology. With this technology, Lycos maintains and continually updates a catalogue of the web, keyed off of the most common words appearing a web home page. Clearly, we have a long way to go in developing search engines to solve the resource discovery problem. More work needs to be done in integrating consumer search behaviour with agent technology. Thus the web architecture that facilities electronic commerce by providing set of tools can make interactive applications come to life.

Intranets and Supply Chain Management

Effective management of inter firm linkages in the supply chain can add value in various ways—lower cost, improved quality, faster speed delivery, improved support and strong competitive positioning. Intranet based electronic commerce technology and supply chain functions are becoming interviewed as the walls between functional structures such as purchasing, personnel and production are dismantled. The focus is on the optimization of integrated processes that extend while taking account of the wishes of the customers. For firms, the Intranet environment offers the promise of a new approach to better address certain business requirements. We see that Intranet computing as highly complementary to the activities on the Public Intranet. What are the Intranet's implications and benefits in managing supply chains.

Intranets and Customer Asset Management

The role of Customer Service Operations has changed greatly. The Customer service function has grown more competitive and information technology needed to support it

Companies are depending more on customer service operations. Electronic Commerce technology in particular data mining can have a significant role in marketing strategy.

Electronic Payment Systems

Consumer acceptance of electronic payment initiative may be slow. It took consumers twenty five years to get accustomed to using ATMs. The introduction of credit cards was down played by arguments that they could never replace cash, no retailers would accept them. It is prudent to examine legal and business issue that affect the pace of change in payment system—customers would take to a paperless and cashless world. Infact, banks have strategic plans that anticipate the future of electronic banking. The challenge for the banking industry lies in creating the right atmosphere and providing the right incentives for consumers to use personal computers regularly for banking and in making sure that they provide attractive and affordable services. To ensure security on the Internet, several methods have been developed. They include firewalls for perimeter security authentification of users and servers, encryption and data integrity. Transaction security is critical information transmitted over the Internet. Online firms must take additional precautions to prevent security breaches. To protect consumer information they must maintain physical security of their servers and control access to software passwords and private keys. Techniques such as secret and public key encryption and digital signatures play a crucial role in developing consumer confidence in electronic commerce.

Internet too can be Destroyed

Researchers at Notre Dame University in the U.S. have found that by taking out the networking with the most highly connected nodes (mother computers forming the cross roads through which Internet data travel, cyber terrorists can actually dismantle the behemoth. The average performance of the Internet surfers by a factor of two if just one percent of the most connected nodes gone, the Internet fragments into small islands of disconnected domains. There are a couple of similar hubs that are crucial to the Internet. If the internal hubs are taken out simultaneously there would be a serious problem.

Electronic Commerce is the ideal starting point for modern managers and professionals/users who want to keep abreast of the latest trends and issues in management practices affected by information technology. New Economy, New Tools, New rules and streamlining of interactions, products and payments from customers to companies and from companies to suppliers is causing an earthquake in many boardrooms. Managers are being forced to re-examine Electronic Commerce and payment systems interface.

Electronic payment is an integral part of electronic commerce. Electronic payment is a financial exchange that takes place on line between buyers and sellers. The content of this exchange is some from of digital financial instrument (such as encrypted credit

card numbers, electronic checks or digital cash that is backed by a bank or by legal vendor. Factors stimulating interest among financial institutions in electronic payments are:

Increasing on-line commerce,
Reduced operational and processing costs, and
Decreasing technology costs.

The desire to reduce cost is the major reason for the increase in electronic payments. Cash and checks are expensive to process and banks seek less costly alternatives.

The main issue in electronic commerce revolves around how consumers would pay businesses on line for various products and services. The solution proposed to the online payment have been adhoc at best. In one of the methods by Cyber Cash, users install client software packages known as Wallets on their browsers. This software communicate with electronic cash registers that run on merchants web servers.

Objectives

The objectives of Online Commerce is to develop a small set of payment methods that are widely used by consumers and widely accepted by merchants and banks. Electronic funds transfer systems have shortened the time of payment instruction transfer between banks and in the process have reduced the float. EFT have not changed the fundamental structure of the payment system. Many payment innovations aim at minimizing banking costs such as reserve requirements, speeding up check clearing and minimizing fraud. Innovations in electronic commerce aim to affect the way consumers, deal with payments and in the right direction of a real time electronic transmission, clearing and settlement system. Several innovations help to simplify payments such as:

— Innovations affecting consumers—credit and detect cards, ATMs, stored value cards and electronic banking.
— Innovations—on line commerce—digital cash, electronic checks, smart cards and encrypted credit cards.
— Innovations affecting companies—the payment mechanism that banks provide to customers such as inter bank transfer through auto marked clearing houses.
— To understand how electronic payments fit into the shopping experience—let us examine the process in details.

The consumer browses for items and selects items to be purchased. The merchant presents the consumer with an order from containing the list of items their prices which include shipping, handling and taxes. This order form may be delivered from the merchants server to the consumer with the ability to negotiate pricing. The consumer selects the means of payment which include digital cash, electronic checks or credit cards. The consumer ends the merchant a completed order and a means of payment.

The merchant requests payment authorization from the consumer bank. The merchant sends the customer confirmation of the order, shipment and payment. The merchant requests payments from Consumer, financial institution. The bank authenticates the individual and disburses funds to the merchant. The merchant delivers the purchased product upon customer verification off line payments depend on:

(a) The V2 an sacking parties—buyer and seller be in each other's physical presence.

(b) Delay in the payment process for defection ofraud, over-draft, other problems to be identified and corrected.

Traditional payment methods do not work on line due to:

(a) Lack of convenience

(b) Lack of Security

(c) Lack of coverage

(d) Lack of eligibility

(e) Lack of support for micro transaction

To suit the needs of electronic commerce, modern companies are developing new forms of financial instruments such as digital cash, electronic money and checks. The main element is the notion of an electronic token for supporting micro payments. Micro payments or small free transactions are the driver behind.

Digital Economy

There are three ways in which consumers pay for their goods, cash, cheques and credit. There are other forms of payments such as money orders, bank cheques, debit cards and travellers cheques. In addition, companies use line of credit, purchase orders and other methods to pay for goods. A few basic approaches for deploying electronic cash, electronic cheques and electronic credit. There are certain banking and other regulations pertaining to handling electronic payments over the Internet. Electronic cash, e-cash, digital money or digital cash provides the means to transfer money between parties over a network such as the Internet. Electronic cash satisfied general properties of digital money such as Independence, Anonymity, Transferability, Diversibility, non-sensibility, secure storage. There is an electronic cash issuing bank that generates electronic cash. Such a bank or institution an e-mint. The e-mint signs being the electronic cash based on the funds provided to it by the customer.

Electronic Cash System

The main elements for an electronic cash system under which cash transactions take place such as:

Obtaining Electronic Cash
Purchasing with Electronic Cash
Redeeming cash by the Merchant

Request to obtain electronic cash, transfer money send Electronic Cash, Deliver goods, redeem electronic cash, and credit Merchant Account.

Thus the exchange of electronic cash between a customer and a merchant. However, a similar scheme can be devised to transfer money between two individuals or institutions such as banks, universities or other businesses.

Electronic Cheque System

In case of electronic cash, five parties—the customer's bank the clearing house to process cheques among different banks. The functions for a clearing house may be handled by a separate entity or by an existing banking system. The consumer uses a web browser that has access to various web servers over the Internet. The consumer views various shopping malls and store fronts at the browser. The browser has provisions for displaying the electronic cheque formats. The banks process a complete electronic cheque transaction consists of several steps. The consumer makes a purchase. In the second phase, the merchant sends the electronic cheques to its bank for redemption. In the third place, the merchants, bank approaches the clearing house or the consumer's bank to cash the electronic cheques.

Security Schemes for Electronic Cheques

The security requirements for electronic cheques consists of authenticating the electronic cheque supplying the originator's public key to the receiver and securely storing the originator's private key. The electronic cheque consists of a document that is signed by the consumer's, private key. The receiver (the merchant or the merchant bank uses the payer's public key to decrypt the digital signature. This assures the receiver that the sender indeed signed. Additionally, the electronic cheque may require the digital signature of the originator's bank. The receiver can validate the authenticity of the originator's bank by using the public key of the originator's bank.

To avoid fraud, the consumer's private key needs to be securely stored and made available to the consumer. This can be achieved by providing a smart card that the consumer can carry.

Electronic cheque can be issued without needing to fillout, mail or deliver cheques. It saves time and lead to reduced paper handling cost as well as reduction in bounced cheques.

However, there are privacy issues to be addressed for e-cheques. The banking system is obliged to document the details of every transaction passing through the system

While maintaining such records, the bank needs to protect the payee's privacy and not to divulge the details of banking transactions.

Electronic Credit

Electronic credit card number and expiration date can be prevented from disclosure to the merchant. The electronic credit system can be designed to obtain instant payments to the merchants from credit card sales.

Smart Cards are credit cards that can be carried in a Wallet. These cards have a processor and memory on the card. In this way, digital cash or private keys can be stored on the cards and can be carried around by the consumer.

SWIFT

Swift the society for world wide Inter Bank Financial Telecommunications, to provide a secure means of facility along international standards for all users of this international messaging and payment systems. Other inter bank transfer applications include BITS (Australia) and KITS (New Zealand). Business reply on electronic payment systems provided by the major banks to accept payment. The system provides with an efficient and secure means of collecting payments, transferring value and managing cash flows. Many organizations have integrated these payments systems into their own memory, cash management, accounts payable and payment approval systems and processes. The technology requires to generate and format a transaction on a PC or similar device. Lack of customer and service provider confidence in the security and integrity of these systems have been key agents in the slow offering and take up of these services. Security tokens, authentication techniques, access controls, confidentially measures and data transmission controls are all important considerations for electronic payment systems. An understanding of the key components, risks, control features and workable procedures must be obtained before an audit can make recommendations to management about safeguard assets and payment in a cost effective and secure way.

Banking Network

The banking network consists of several components. There is a bank that processes the on line financial transactions for the given merchant. The bank maintains the accounts for the merchant authorizes and processes the payments. The merchants bank also maintain a link with the consumer's bank for verifying the transactions. The link between the merchant and its bank is often real time so as to allow on line authorization of consumer payments. The consumer's bank typically has an off line to the consumer such as via postal mark or e-mail.

Steps are executed a complete transaction

— The consumer accesses the shopping mall and selects a shop for purchasing certain items.

— The shopping mall server accesses the merchant system for the selected shop
— The merchant system presents the store's home page to the consumer.
— The consumer selects the desired goods, interacts with the merchants and makes the payment.
— The merchant system accesses its bank for authorization of the consumer payment.
— The merchant system informs the consumer of the money transfer through mail.

Banking System in on-line Commerce

Banks have a variety of technological means towards on-line banking programmes. The reach and delivery capability of computer networks such as the Internet far exceeds any proprietary bank network ever built and makes it continually easier for customers to manage their money any where, anytime. Banks understand the need to separate the content (financial product) from the distribution channel. The traditional model of integrated delivery has out lived its usefulness and needs to be reconsidered. Banks recognized the potential benefits of early entry into what will inevitable be an important but crowded market. The challenge for the banking industry lies in creating the right atmosphere and providing the right incentives for consumers to use personal computers regularly for banking and in making sure they provide attractive and affordable services

Electronic Payment System

Electronic Payment Systems are widely used in commerce including whole sale payments, wire transfers recurring bill payments, the auto marked clearing house, electronic draft capture and electronic check payment. The emerging area of cashless payment and the various payment options that are currently being examined. These payment options include e-cash, electronic checks, smart cards and electronic purses. A world with electronic payment system will be different from today's world. New opportunities would arise for consumers, banks and others. If record keeping were part of the payment instrument, several innovative uses could be visualized. Consumers could tally up every thing they spent with little effort. They could do realistic expense analysis and even negotiate with merchants based on volume. Even small merchants could know who utilizes their stores and how much they spend. Possible benefits to banks and other firm would be a better understanding of the financial behaviour of their customers. Governments might monitor prices, inflation and velocity of money more accurately and more quickly than today. The challenge for the future is to come to the right vision of how electronic payment systems will arugment or replace paper based payment method. The next step is know out how to make a profit on that vision

Network Security

There is an increasing demand for efficient collection, dissemination and processing of

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information due to various economic and market forces. Electronic convergence makes use of today, technology to solve business communication, computer intensive business problems. Today, Computer systems, linked by global networks face a variety of security to create that result in significant information and financial losses. Threats vary considerably from threats to system availability from malicious hackers attempting to crash system to data or transaction integrity threats. An understanding of the types of threats in today's computing environment can assist a security manager in selecting appropriate cost effective controls to protect valuable information resources. An overview of many of today's common threats will be useful to organization studying their own threat environments with a view toward developing solutions specific to their organization.

The Network as a Network Infrastructure

The Internet has transformed the way network infrastructure are conceptualized and created. In the past, network infrastructures were built around highly structured architectures that were planned, created and operated by Government Velcos and big Corporation. The basic infrastructure flowed top down from formal meetings and standards that purported to chart the future requirements for decades to come. Interface between Electronic Commerce and payment systems, we see electronic commerce as the integration of net work infrastructure, data management and security services to allow business applications within different organizations to interchange information within the context of old business processes automatically. Communication services transfer the information from the originator to the receipt. Data Management services define the interchange format of the information. Security services authenticate the source of information received by the receipt, prevent disclosure of the information to unauthorized users and verify that the intended receipt received.

Electronic Commerce applications are being built on a foundation of global hypertext. Developers believe that hypertext is just another user interface approach. Dislinked hyper text systems such as the worldwide web are computing hybrids that span traditional boundaries. The web is a networking technology plus an information management method that provides a novel way of directly accessing and manipulating information. It is an interface that integrates security and payment schemes and can be used for on-line buying and selling.

Thus the Web provides a totally different and unique method of accessing information. Whereas traditional data bases have some structure to them, a hypertext data base has no regular structure. The user is free to explore and assimilate information in different ways, an immense flexibility that is making electronic commerce a reality. Globalization has put increased pressure on companies to operate at entirely new levels of effective companies have to deliver more by fundamentally rethinking them existing processes. This can be done with the use of information technology for more effective

decision making. Simply replacing paper documents by the electronic ones would not bring benefits to companies till right methodologies precede the implementation of EC. The challenge for the future is to come to the right vision of how electronic payment systems will augment or replace paper based methods.

Legal and Business Principles Relevant to Electronic Commerce

A combination of certain tools including agreements-legal and business practices and policies, security technologies along with supplied for and refinement of these tools through legal reform is essential to the success of electronic commerce. Legal requirements and business policies clarify and implement reasonable security measures with- out sacrificing needed flexibility, careful planning and rigorous attention to these issues would contribute to a viable electronic commerce environment in the future. The inherent link between security and legal efficacy is to be adequately appreciated. The security of electronic messages and records is not only a business requirement but also a legal necessity. The application of information security technologies combined with an ever increasing commitment to legal and business reforms would result in more sophisticated and comprehensively legal and practices and ultimately in greater efficiency and benefit from electronic commerce.

Security technology may secure the routes of Internet communication but it does not protect consumer from people with whom they might choose to do business. The situation is analogous to telling an unknown person the credit card number over the telephone the consumer may secure in knowing and works for the company they wish to buy from but to conduct business the consumer must be willing to take the extra step and trust the person and the company. Like wise, online firms take additional precautions to prevent security breaches to protect consumer information, they must maintain physical security of their servers and control access to software passwords and private keys. Techniques such as secret and public key encryption and digital signature play a crucial role in developing consumer confidence in electronic commerce.

Electronic Payment Systems

Various types of electronic payment schemes that are emerging and consumer acceptance of electronic payment initiatives may be slow. Many people are skeptical about deposits and it takes consumer years to get accustomed to using ATMS. The use of credit cards could never replace cash and consumer responsive to ATM was not so positive consumers express indifference to something new simply because they do not comprehend what it is and what it can do for them. In order to have sense of proportion of what is possible and it is prudent to examine the key consumer, legal and business issues that affect the pace of payment system change. However, the question revolves around consumer behaviour and acceptance how customers take to a paperless and cashless world.

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Electronic Banking

New technology is enabling banks to provide the convenience of any where, any time banking to demanding customers. The earlier brick and mortor branch is insufficient, new technology is taking banks to the homes or offices, 24 hours a day, 365 days a year through ATMS, telephones and PCS. The financial supply chain has under gone a fundamental strategic change. Funds could be transferred through mail transfers by issuing demand drafts physically transported from its place of issuance to its place of payment. Realization of such payments used to take time. The new trend of Non Branch Service Delivery in Banking has strated with electronic payment services. It strated with Electronic Funds Transfers (EFT). Then credit cards, ATMS and smart cards etc. with the advance of computing technology, telephone banking and its integration has become a powerful medium of delivering banking services. The new technology has broken the paradigm of branch banking, it can be done from home/office using the PC or the telephone or with satellite communication links. The networking has allowed the customer to do banking at any branch. This is the new flexible financial supply chain to achieve the objectives of competitive supply chain management that are being used by the banking Industry in the form of non-Branch Service Delivery.

Effectiveness of Electronic Banking System depends on the following factors:

- Communication Technology being in its infancy and being inadequate for global coverage-customers and banks could not communicate internationally within their own organization or with each other.
- Companies and banks have incompatible systems.
- Computer manufacturers views on the development of technology standards which would permit data exchange.
- However, several new innovations help to simply consumer payments, consumers credit and debit cards and automated teller machines. Innovations enabling electronic commerce digital cash, cheques, smart cards and encrypted credit cards and EDI and internal technologies innovation affecting companies such as inter-bank transfer through automated clearing houses that allow companies to pay dividends to their shareholders Technological advances have made home, office, telephone banking more effective and efficient as a means of selling and delivering products and these channels are gaining more acceptance among customers with a view to improve their competitive position with respect to the key success factor.

New Scenario

The new scenario for the buyer and the seller to be a system where the payments

settlement happens at the point of sale itself when the goods and services are exchanging hands. This is what electronic banking achieves. Through electronic data networks, it makes the payment on line, at the sametime as the actual exchange of goods and services takes place. From distribution point of view, it brings the bank closer to the consumer at a telephone call or at his PC. It is this way that is being taken by the foreign banks and the private banks to attract their customers, extend the reach of the target market for client account acquisition and service delivery capabilities beyond the branch for sales and services of their customers. The re-engineered supply chain of Banks has a few components.

- Branch acquisition
- Remote acquisition
- Branch servicing
- Remote servicing
- Centralized operations facility and
- Vendors and suppliers

Branch acquisition or sales are restricted to customers who walk in to the branch to open an account or apply for a loan. Once the customer interaction is over, the documentation for setting up a new account or disbursing a loan moves to the centralized processing site which takes over the rest of the processing.

The branches maintain some infrastructure to service customers to transact on their account transactions which do not require customer interaction are shifted to a centralized facility which can service multiple branches.

Remote servicing is the area where technology is playing a big role in reducing transaction cost and giving customer convenience at the same time. ATM at shopping centers, airports, railways stations, full service phone banking and PC/Internet banking are revolutionizing the way customers are using banking services.

Centralized operations facility services, multiple branches and does all back room driven processing activities. It issues cheques books, ATM cards, fixed deposit receipts or loan disbursal payments, processing clearing checks and trade finance.

Vendors and suppliers need most of low value added activities like statement printing and mailing cheque book printing and mailing, ATM card embossing even some low risk data entry jobs are out sourced to external vendors and suppliers with a view to bring costs down.

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Exhibit showing Distribution Model

Customers

↓

Distribution

↓

Remote acquisition	*Remote servicing*	*Branch servicing*	*Branch acquisition*
Direct associate visiting Banker Extension Counter Direct mail	Cash & delivery strategic tie ups ATM/ Phone Banking PC/Internet Banking	Teller function Remittance trade finance	Walk in

→ Centralized Bankroom ←

Cheque book/ATM card state Issuance fixed deposit issuance loan disbursal cheque

vendors/suppliers

Computerization in Indian Banks

The encouraging sign is that bank computerization has acquired a substantial momentum in banks in India. The foreign banks in India have computerized since 1980 and most of the branches are connected to each other. The private banks are fairly new and most have strated their operations with computerized networks. Even in the public sector banks, bank computerization has come a long way. The unions of these banks have agreed to the introduction of electronic funds transfer, computerization of major clearing houses, setting up of communications networks to connect the branches. PCS have become a common site in bank branches.

Issues in Electronic Banking

The key issue is to have a judicious mix of both direct distribution and indirect distribution to increase breadth of geographical coverage while keeping the costs low. Customer's comfort with new technology is increasing, it is yet to reach a critical mass in India while a large proportion of the population does not understand technology at

all, white electronic banking and the Internet do off exciting opportunities for payment mechanism, there are challenges that have to be addressed without which this form of banking cannot make significant progress. On line banking would meet its full potential when the following issues are addressed properly.

(a) Consumer protection from fraud,

(b) Transaction privacy,

(c) Infrastructure issues of telecommunications and bandwidth,

(d) Comfort with technology and proliferations of PCS.

(e) Legal recognition of electronic contracts.

Electronic fraud is a scourage in the west and is also the rise in India. Unless electronic message transmission can be safeguarded from unauthorized access through hacking, people continue to be hesitant about using electronic banking for fear of fraud. In India, cash transactions are popular because of the anonymity they provide, as they leave no paper trail. A part of Indian economy is the black money, where transactions are in cash and there are no audit trails. This is a factor that continue to hinder wider acceptance of electronic banking because electronic payments always a paper trail. Telecommunication infrastructure is robust and this requires large scale area network connectivity, India has a long way to go. The average Indian consumer is still PC illiterate and prefers the human touch over technology driven service. Information technology is still a long way when PCS become like TVS in every household. The consumer behaviour toward technology would take some more time to change till such time, electronic banking would remain restricted to the younger upwardly mobile consumer only. There are some challenges that stand in the way of electronic banking gaining momentum however some progress has been made with the passing of the cyber laws in the form of Information Technology Act, 2000 passed by the Indian Parliament and now a relevant question is about its implementation effectively. Despite the low diffusion off technology in India, the momentum of electronic banking has picked up led by the foreign banks and the Indian private banks, it is expected to accelerate in the near future.

Internet and Banking

Banks have a variety of technological means to initiate online banking programmes without incurring the investments needed to develop their own systems. The delivery capability of computer networks such as the Internet far exceeds any proprietary bank network ever built and makes it easier for customers to manage their money anytime and anywhere. Internet and Mobile phone usage have been growing at a fast pace which would facilitate acceptance of electronic banking. There is need for Banks to separate the content *i.e.* financial Product from the distribution channel *i.e.* the branch. Clearly the traditional model of integrated delivery has out-lived its usefulness and needs to be modified. The Indian consumer at large is still not completely comfortable

with technological advances as a way to have electronic banking. The younger generation have been using computers which would be the consumer of the future. It would take time before this technology becomes the technology for the masses in India. While electronic banking and the Internet do offer exciting opportunities for payment mechanism, there are many challenges that have to be satisfactorily addressed without which electronic banking can not make progress unless.

- Internet access becomes more wide spread in India.
- Security aspects of transactions over the Internet improve.
- Cyber laws are implemented effectively to give legal sanctions and validity to electronic signatures.
- Gaining experience through these features would help to sharper on skills and reduce the risks what matters most is a right mind focused on results and staying motivate and committed. We need to be alert to avoid getting trapped in the vicious circle well informed persons have clarity in their head as well as their heart. If seeds of good communication are sowed, trees of benefits would definitely accure. Experiences are widely shared, however, the willingness and commitment to such a system to achieve the best results. Thus there is a great need for—
- A clear vision of tomorrow, banking environment and the challenge in creating the right atmosphere and providing the incentives for consumers to use personal computers for banking and in making sure that they provide attractive and affordable value added services.

CYBER LAWS

The aggregation of Intranets (closed circuits of interlinked or networked computers) Internet (world-wide network of computers linked through various telecommunication links) and the wordwide web (the networking of computers world wide with ready access to each others data through communication protocols and hyper text marked up language links) is dubbed as Cyber Space. Cyber Laws, are those laws which have been adapted or re-interpreted to govern or apply to transactions or interactions in Cyber Space. Cyber laws also cover those special enactments which are specially designed to govern or apply to Cyber Space for example, the Electronic Commerce Act proposed by the Government of India or the Uniform Computer Information Transaction Act recently approved in the U.S. for adoption by the various US States.

Contracts

The Electronic Commerce Bill does promise to make electronic contracts feasible. However, we need to learn from the experiences in other countries and to examine how our current contract laws could be applied to such contracts.

Privacy

In an issue titled, The End of Privacy states that "the volume of data recorded about people will continue to expand dramatically. Disputes about privacy will become more bitter. Attempts to restrain the surveillance society through new laws will intensify"...."People will have to start assuming that they simply have no privacy. This will constitute one of the greatest social changes of modern times".

The U.K. Data Protection Act of 1984, stipulates stringent rules for registration of personal data users and for regulation of the purpose, sources and use of such personal data.

In the USA, the tussle is between the desire for personal privacy and the fundamental and cherished principle of free flow of information.

Domain Names

Recently, the Internet Corporation for Assigned Names and Numbers (ICANN) adopted a proposal for resolving domain disputes which will require mandatory arbitration of all such disputes involving trademarks, with the loser paying costs. This will become effective when adopted by Network solutions Inc., and other domain registrars and is incorporated in the appropriate contracts with domain name registrants.

Patents

The most profound and significant implications, however, appear to be in the area of patents. The spate of such inventions which relate to the exploding world of electronic commerce and the value of Intellectual property is best exemplified by a company called Digital Walker which has about 12 such patents and over 240 patent applications in this area. The company is valued at several billion dollars within less than 2 years of formation and about half of its total staff of only 25 are lawyers with patent experience.

The Role of ICSI

Company Secretaries, with their broad legal, financial and general skills are best equipped to take India and its entrepreneurs into the Internet world of commerce.

Here is what the ICSO can do:

- Influence and help formulate a national e-commerce framework;
- Create websites for exchange of views and debates on e-commerce laws;
- Help create the 'soft' infrastructure needed; for example, insurance and liability advisories, funding and IPOs, venture capital networks, legal advisories etc.;
- Permit and encourage foreign linkages for practising company Secretaries so that Indian firms can team more rapidly and create a global network;

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Information

- Introduce specialized streams in the course for students in e-commerce laws and intellectual property laws;
- Preview and seek changes in other laws which fetter e-commerce; set upon working group for this.

Significance of Global E-Commerce

The industrial revolution was perhaps the biggest change that business ever faced. In the beginning, we did everything manually, recording them on paper. In order to speed up the work we computerized it. Then we set up networks to share information pertaining to work. In our simple model of the business universe, the network first appeared within the organization. Interactions of the network with the outside world were almost zero. For any data to get added to or get out of the system, manual intervention was required. Some one had to take a printout and manually re-enter the data into a different network or copy it on to a floppy and carry it across. Hence interactions across the interface are often time consuming and require elaborate process and protocols. But in today's information age, this interface is rapidly crumbling and is being replaced by new mechanisms that make it easier to transfer information across the boundary.

Electronic commerce is a modern business methodology that addresses the needs of organizations, merchants and consumers to cut costs while improving the quality of goods and services and increase the speed of service delivery using computer network to search and retrieve information. E-commerce is associated with the buying and selling of information, products and services via computer networks. In other words, it is a means of transacting business electronically and in many cases over the Internet. It involves a composite of technologies, processes and business strategies that foster the instant exchange of information within and between organizations, buyers and sellers. Information Super-high-way (I-way) will transform information, transport technology for electronic commerce applications and provide an economic windfall. E-commerce is well suited to facilitate the current re-engineering of business processes occurring at many firms.

Objectives

Objectives are: reduced costs, lower product cycle times, faster customer response and improved service quality by reducing paper work and increasing automation. Today the emphasis has shifted from the narrow focus to the invention of entirely new business applications for reaching and getting close to the customer. New types of information based business such as on-line advertising and marketing, on-line order taking and on-line customer service which reduce costs in managing orders interacting with a wide range of suppliers and trading partners. E-commerce facilitates information of new types of information based products such as interactive games, electronic books and

Com ication

information on demand. E-commerce results in improved efficiency in finding and interacting with customers in communicating with trading partners and in developing new products and markets. Key element of e-commerce is information processing. Bottlenecks are long as transportation distances, customs regulations, language barriers etc. and coordinating them through software via the I-way can reduce the complexity.

E-commerce is a new way of conducting, managing and executing business transactions using computer and telecommunication networks. E-commerce is expected to improve the productivity and competitiveness of participating business by providing unprecedented access to an on-line global market place with millions of customers and thousands of products and services. Another objective is to provide participating companies with time efficient means for working with customers, suppliers and development partners.

E-commerce is the tool that leads to *'enterprise integration'* for a company for an industry and ultimately for the vast network of small business, government agencies, large corporations and independent businessmen.

An e-commerce site will enable one to extend the reach of his business by bringing in customers from previously untouched regions without incurring additional expensive advertisement compaigns. E-commerce has definitely not left India untouched though in India it still has a long way to go to before it matches international standards.

Convergence

Many companies are pooling their resources and talents through alliance and mergers with other companies to make the electronic market place a reality. The term e-commerce has become irrevocably linked with the idea of convergence of industries centred on information. "Convergence" broadly defined as the meddling of consumer electronics, television, publishing telecommunications and computers for the purpose of facilitating new forms of information based commerce. Impact of convergence broadly includes- Widening of the portfolios and building of existing customer relationships, creating opportunity and pressure to enhance delivery capability through services or products resulting in additions of more and more products supplementing each other in its, acquiring of small companies with innovative technologies by bigger ones to fill gaps in their product range, formation of strategic alliance in order to deliver complete integrated solutions etc. the whole process of convergency has been considerably slow because implementation requires large initial investment.

There are three compelling drivers that favour the emergency of converged networks—Capital savings, Operational savings and tariff savings. Seeing the recent market trend, service providers and equipment vendors are planning convergence, otherwise they are at risk of being sidelined and over taken in the race. An important trend in the IT industry is the speedy change in technology. Convergence will be cost-

productive for corporate with geographically spread offices. Automating the flow of information and the process of decision-making help businesses substantially improve their time cycles, reduce inventories and generally improve the ROI.

Multimedia content can be considered both as fuel and traffic for electronic commerce applications. Digital data in more than one format, such as the combination of text, audio, video and graphics in a computer file/document. Business professionals are well aware that more than 90% of the information that firms use for business operations and decision-making lives outside the "traditional" database systems. The goal of multimedia is to increase the utility of all information through the processing and distribution of new forms such as images, audio and video. Electronic commerce requires *robust servers* to store and distribute large amounts of digital content to consumers. These multimedia storage servers are large information warehouses capable of handling various content ranging from books, newspapers, advertisement catalogues, movies, games and x-ray images. Stready advance in digital memory technology are making mass storage devices technologically feasible and increasingly cost effective.

Internet

Internet is a global resource inter-connecting millions of users based on a standard set of protocols—a mutually agreed upon method of communication between parties. It is slowly taking over the world of communication in a big way. Of course, the ideas different people have about the Internet may differ but infact it could be termed as the most powerful and important technological advancement. The dream of getting the whole world connected in such a way that information of any kind can be accessed and retrieved from any place has been ultimately realized through Internet. The FTTP (File transfer protocol) is a tool that allows transfer of files between Computers, which in most cases would be connected over the Internet. With Internet, people no longer need to depend on radios as the only source to tune into its individual stations. This is because web allows its audience to listen live from various broadcasting stations. Internet can be leveraged to provide better customer service and tremendous time and cost savings.

Corporations see the internet as a great way to put their sales literature on line. Number of web sites increased from zero in 1993 to over 4 million today which was better utilized by companies like Yahoo. Many companies are scrambling to connect electrical and electronic gadgets to the Internet. Computer geeks are brimming with new ideas and researchers are competing to be the first to introduce new products or solutions that can provide such capabilities. Examples include kiosks, set-top boxes, technologies that enable mobile phones etc. People to talk to each other via desktops/laptops with internet connections. One day every electronic gadget that one carries or encounters in public will possibly have Internet connection capabilities.

Now one could directly place an order on his supplier's network without any

intervening paper work. This enables to plan better, reduce inventories, improve turnaround time etc. this is widely known as EDI (Electronic Data Interchange) system a fast and dependable way to deliver electronic transactions by computer to computer communication which is cost wise prohibitive particularly when a client and his suppliers are operating far apart.

Challenges

A challenge facing e-trade is the absence of a clear-cut regulatory framework worldwide. India, with its complex regulatory framework needs to define transparent rules for e-commerce to keep pace with global growth. A number of issues like *taxation, tariffs, data protection, authentication, privacy* and *copyrights* need to be reviewed from the e-perspective. This is a new challenge for lawmakers. E-trade dissolves boundaries to bring in paperless dealings. Indistinct boundaries make it difficult to impose customs other taxes on goods and services traded over the Net. If orders are booked on the Net but there is physical delivery of goods, taxation is possible. But how to tax a product that has been delivered over the Net-software consultancy services or other information?

Countries looking for alternative sources of revenues are planning to levy tariffs on global e-commerce. While trading on the Net, the seller needs to be sure that his intellectual property rights are protected. There are treaties that establish international norms for the protection of copyrights, most notably the Berne Convention for the Protection of Literary and Artistic Works.

Advertising

Advertising through television is costly compared to web advertising. Moreover, online advertising is more likely to be noticed than television advertising. Web users actively use the medium as opposed to passively receiving it. Few people actually pay full attention to television shows, not even two-thirds of the viewers of television's most popular shows are paying "ful attention" to the programming. Web surfers can not navigate the web without high concentration and attention levels. Both are required for advertising to get noticed, remembered and ultimately acted upon.

Some of the concerns that need to be addressed relate to the area of taxation of goods and services traded via e-commerce. Tax treatment on such electronically traded goods under various tax levies such as customs, excises, sales tax, service tax etc. create problems. Although it is proposed to free all electronic transmissions from customs duties, what about taking on such products depending on its mode of delivery.

Authenticity of transaction on the electronic media is another concern. The identify on the Net is often anonymous, but how important is it to disclose the real identify? Another concern with authenticity is that there is no paper that holds the stamp or signature of the parties involved during order, delivery or payment. Is such a transaction a contract? Does it legally bind the parties who electronically enter into such an

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agreement? Does not meeting the terms of the contract amount to breach of contract as stated in the Contract Act?

Issues like a customer denying that he placed an order on line requires sophisticated authentication mechanisms. It is not clear whether an authentication mechanism can be recognized by an Indian Court as a valid evidence. Efficient legal mechanisms for such systems are required to succeed otherwise customers in developing countries can get badly hurt.

Privacy is another important issue on the Net. Should an organization share the personal information that it collected beyond the extent that it was collected for? ECLIP (Electronic Commerce Legal Issues platform) a committee set up by the European Union has laid down a legislative framework for privacy on the Net.

Electronic commerce is an enterprise issue. Business today are faced with more competitive pressures than ever. They are under unrelenting pressure to reduce costs remaining responsive to both customers and suppliers.

This pressure has driven business to recognize the need to automate and coordinate the flow of information between front-end and back-end of the business. Most of the organizations are integrating their existing enterprise applications with electronic commerce transaction capabilities to manage the flow of business transactions internally and externally across their customer and supply chain, become more customer centric and, become what has been identified as the "next generation enterprise" *i.e.* one that relates to its customers, suppliers and partners *via* electronic means. Therefore, enterprises today need to re-engineer themselves and their processes around electronic commerce in a similar fashion to the way that many enterprises have been re-engineering themselves around Enterprise Resource Planning (ERP) in recent years. Supply Chain Management and Customer Relationship Management are two electronic commerce related initiatives that have to be taken care of.

Banks and financial institutions stand to benefit enormously from e-commerce in the form of reduced transaction costs, improved customer service and loyalty and better customer information. They also get benefited from helping business handling payments and financial transactions by e-commerce.

Global Trust Bank Ltd. has entered into a strategic alliance with Infosys Technologies Ltd.—to facilitate its foray into electronic commerce by deploying Bank Away-a versatile platform for internet banking. A powerful electronic commerce platform that enables banks to provide an integrated financial services offering to their customers the one-click access to all their bank accounts, trade finance, bill payments and investments on line shopping and more.

GTB plans to initially offer its customers a single view of all their accounts at any of the bank's branches. Customers will be able to know about their accounts, request for

transfer of funds between their accounts, cheque books and demand drafts. GTB plan to offer value added services such as access to their trade bills, letters of credit and bank guarantees. One particularly important area for cost reduction is in the supply chain. Taking expenses out of the supply chain could be achieved through e-commerce.

Secure Electronic Records and Signatures are afforded higher evidentiary presumptions to provide parties engaged in electronic commerce assurance that their transactions are enforceable. The security procedure must satisfy four criteria such as uniqueness, objective identification, reliability and linkage to record signer. It is presumed that the secure electronic signature affixed to an electronic record is the signature of the person identified as a signer. If there is evidence that the person whose signature affixed was the victim of mistake, misrepresentation, duress or other invalidating cause, the record may be denied legal effect, but the burden of raising these issues is on the person denying the legal effect of the record. These presumptions apply only in a civil dispute and not in a criminal matter.

A person relying on digital signatures assumes the risk that the signature is invalid in circumstances where there is a questionable digital signature one that cannot be verified because of several reasons such as an error by the signer or faulty digital signature system.

Whether digital signatures on documents can be legally considered binding on the parties to the contract? How to avoid computer related frauds such as hacking, malicious falsification or erasure of data, software thefts, software attacks? Does this require self-regulation by business community or through governmental intervention? Hence timely prevention and detection of Cyber crimes and frauds has to be considered. It is practically difficult even for the regulatory authorities in the developed countries to track down Cyber Fraud relating to securities, such as frauds relating to purchase of securities through the internet money laundering etc. Instituting suitable legal changes and creating enforcement mechanisms are perquisites followed by accountability by creating suitable technological standards for automatic monitoring of electronic payments and transactions. Detection of Cyber frauds and prosecution would also involve police officials as well as prosecutors in advance technology apart from greater international cooperation between enforcement agencies. The International chamber of Commerce believes that restrictions on cryptography failed in fighting crime.

The Electronic Commerce Transaction

Electronic transactions are conceptually very similar to traditional (that is, paper based) commercial transactions. Vendors present their products, prices and terms to prospective buyers. Buyers consider their options, negotiate prices and terms (where possible), place orders and make payment. Then, vendors deliver the purchased products. While the precise order of these events and the mechanisms through which they are transacted

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vary, these activities, are in principle, fundamental to both traditional and electronic commerce.

Electronic commerce raises some new and interesting technical and legal challenges. These include:

- Satisfying traditional legal requirements for reduction or agreements to signed documents;
- Applying legal rules of evidence to computer-based information; and
- Interpreting, adapting and complying with many other existing legal standards in the context of electronic transactions.

From a legal perspective, one of the most significant issues in electronic commerce is how to create enforceable digital contracts for the sale of goods and services or how to ensure that digital transaction will be at least as enforceable and valid as a traditional paper-based transaction.

In every business environment, whether transactions are executed in person (face-to-face) or over distance, there are accepted customs and practices that determine, in conjunction with applicable legal rules, the parties, rights and responsibilities. These practices often include controls, such as:

Signatures to evidence agreements;

Time and date-stamping, to provide proof of dispatch, submission, receipt or acceptance; and

In some case, witnesses, notaries or other trusted third parties, to acknowledge and authenticate transactions.

The purpose of these controls is to create the necessary level of certainty in business transactions. Although electronic commerce is expanding rapidly, the development of a corresponding legal and control infrastructure has lagged behind to create viable electronic equivalent to traditional contracting activities, it is necessary to develop legal mechanisms or supportable legal analogues, for the electronic commerce infrastructure. The goal of such mechanisms is to make electronic transactions at least as efficient, secure and legally binding as traditional commercial transactions, without forcing users to negotiate customized terms and conditions.

Creating a Binding Commitment

At the heart of an electronic commerce transaction is the intention-indeed, the critical need-to form a legally binding agreement between the transacting parties.

Consideration

A valid contract also requires that the parties bargain for consideration. Consideration

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may consist of either actual performance, such as delivery of goods or services or payment for them or a return promise. Although electronic commerce may involve novel methods of payment and delivery, as long as a transaction includes a bargain for exchange of adequately commensurate promises or performances, regardless of the manner of performance, the agreement will comply with the consideration requirement.

Statutes of Frauds

Although a valid contract may be established through oral offer and acceptance, courts in the US and certain other jurisdictions typically do not enforce agreements involving various types of consideration unless they satisfy the statute of frauds, the legal requirement that some more or memorandum in writing and signed by the parties must exist for agreements of these types to be enforceable.

Opportunities

The combination of judical acceptance of new technology, the development of trade usage and legislative and administrative enactments suggests that electronically formed agreements will be found to constitute enforceable contracts for most statute of frauds purposes. Until legislation and trade practices become sufficiently uniform and widespread to provide comfort to parties in the jurisdictions in which they contract and otherwise conduct business, parties who wish a greater level of certainty for the time being should execute written trading partner agreements, thus, establishing more concrete rules.

Performance

Once a valid agreement has been reached, it is the duty of all parties to the contract to fulfill their end of the bargain; their efforts (and obligations to do so) are termed performance. Performance by both sides is, of course, what the parties have bargained for obtaining performance is their object in forming the valid and enforceable contract. Thus, when one party fails to meet the legal requirements for satisfactory performance, there is a breach of the contract and non-breaching party may be entitled to certain rights against the breaching party.

Performance in many electronic commerce transactions involves electronic media, especially in the payments process or where the contract is for the provision of on-line services such as access to information or down- load of software. Download of software and the access of billed information resources are examples of performance which can be completed without human mediation (beyond the initial programming).

Compliance

A party's failure to perform completely and strictly in accordance with the terms of an agreement constitutes a breach of contract. Indeed, with respect to the sale of goods, the law has historically embraced the perfect tender rule, a standard entitling a buyer to

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reject goods unless the seller complies strictly with other quality and quantity provisions of a bargain. Nevertheless, where one party has substantially performed in good faith, it would frequently be unfair to force that party to forfeit all of his or her efforts simply because he or she has not fully complied with the contract. Thus, contract law softens the harshness of the perfect tender rule (or exact compliance) in some situations.

Breach

As stated above, a party's failure to perform as agreed results in a breach of contract. This includes both failure to perform according to the terms of the contract once the time for performance has arrived and the refusal to perform even before the time for performance has arrived (termed anticipatory repudiation).

Depending on the nature of the breach, a contract may be void on its face, voidable by the non-breaching party or severable, meaning that certain terms might be voided without affecting the validity of others. (in a case of anticipatory breach, if the non-repudiating party has fulfilled its end of the bargain, the result is a total breach, giving rise to various rights on the part of the non-repudiating party, including the right to terminate the contract and make claims for damages).

Enforcement

One of the fundamental objectives of contract law is to protect a party who accepts promise in a properly formed agreement from injury as a result of a breach by the party who makes the promise. Accordingly, the law affords non-breaching parties various avenues of recourse to enforce the rights under the contract. Of course, a party who has been or stands to be, injured as a result of a breach must be able to prove the injury and the damages that flow there from a court, under the applicable rules of evidence, to enforce these remedies. The liability and damages a party might face for breach of contract and then addresses several rules of evidence that are particularly significant for the enforcement of contracts in electronic commerce situations.

Liability and Damages

A party which breaches an agreement may face various types of liability under contract law. In contracts for the sale of unique goods or other property, a plaintiff is typically entitled to specific performance of the contract by the defendant. An award of specific performance requires a court order demanding the defendant to deliver the goods or services to the plaintiff. Because specific performance is an extraordinary remedy, the capability to accurately identify the person whose specific performance is demanded is very important.

The potential for liability due to statutory provisions for conclusive legal attribution provides additional incentive for electronic commerce participants to employ adequate security measures. In addition to employing information security mechanisms and other controls, techniques for limiting exposure to liability include:

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- Trading partner and legal technical agreements;
- Compliance with recognized procedures, guidelines and practices;
- Audit and control programmes and reviews;
- Technical competence and accreditation;
- Proper human resource management;
- Insurance;
- Enhanced notice and disclosure mechanisms; and
- Legislation and regulation addressing relevant secure electronic commerce issuing.

Rights and remedies are meaningless in the real world unless they can actually be enforced.

However, the parties engaging in electronic commerce should seek to establish their own inter-party presumptions through trading partner or other agreements. For example, the Model EDI Trading Partner Agreement stipulates that certain electronic documents "will be admissible as between the parties to the same extent and under the same conditions as other business records originated and maintained in documentary from". Participation in electronic commerce raises various other legal issues. The list is not meant to be exhaustive, but merely to highlight and introduce some of the most relevant current developments and practices in secure electronic-commerce.

Notice and Conspicuousness

The accommodation of various requirements for legal notice and for conspicuousness via computer has been a source of continuing concern and uncertainty in electronic commerce. Traditional law and legal practice are premised on the use of the mails, the availability of return receipts and other forms of confirmed delivery and on paper documents which permit large font type, capital letters and other conventions to call the recipients' attention to particularly important matters. Unfortunately, current legal standards provide no clear guide in this area.

Intellectual Property

Electronic commerce systems demand greater vigilance to ensure that parties do not violate or infringe upon existing copyrights, trademarks, patents or other intellectual property rights. In addition to copyright and trademark litigation, regarding the context of certain communications, there has been significant legal activity in the trademark area with respect to the use of Internet domain names. Some companies have registered domain names using the trademarks or trade names of their competitors. Network Solutions, Inc. (NSI), the entity that registers.Com domain names, continues to struggle with increasing and novel disputes. In November 1996, an Internet International *Ad*

Hoc Committee (IAHC) was constituted to consider various reforms to the current Internet domain naming system.

Although the courts have not, to date, distilled clear legal rules with respect to various intellectual property issues involving electronic commerce, including domain name disputes, parties have succeeded in obtaining definitive resolutions in certain cases.

Electronic commerce participants should be apprised of developments in the area of computer crimes. Even unsophisticated users have the ability to gain access to, use, misappropriate, alter or destroy information, records or communications, including sensitive payment information or trade secrets.

Legislation and Regulation

Although policy-makers, in general, have not been quick to recognize the potential and inevitability of electronic commerce, certain legislators, regulatory agencies and domestic and international trade and legal organization have increasingly focused attention on electronic commerce law. Much of their efforts remain in the proposal or draft stage, but the trend is encouraging. On global scale, the most significant effort to address legal issues relevant to electronic commerce has been the UN Model law on Electronic Commerce.

UN Model Law on Electronic Commerce

A product of the UN Commission on International Trade Law (UNCITRAL), the Model Law is intended to advance the legal standing of electronic commerce by removing barriers to computer-based trade. Because no jurisdiction to date has adopted the Model Law as its own law, the Model Law is not binding upon any party to electronic commerce (unless its terms are specifically incorporated by reference in a trading payment agreement or other contract). Nonetheless, due to its precatory influence, the Model Law has greatly impacted the law of electronic commerce and the actions of participants in electronic transactions. An outline of the current Model Law is provided in the Model Law is presently incomplete, but is intended "future additional provisions" to be drafted by UNCITRAL. Apart from this Model Law, UNCITRAL's next initiative relating to electronic commerce and communications will most likely be a Model Law or Digital Signatures.

Digital signatures face two challenges in the electronic world. The first challenges is to authenticate the person who is signing the document so as to ensure it is executed by the person by whom it is claimed to be and the other challenges is to authenticate the integrity of the document itself so as to ascertain that it has not been tampered with or cause any alterations to the document to be detected. This will help in preventing forgers and impersonators from 'signing' the document and will prevent a person from unilaterally modifying an executed document without such modification being detected.

Apart from these services, digital signatures also authenticate the origin of the document and with the help of other allied service providers help in authenticating the delivery origination and receipt of the electronic document.

The Information Technology Act, 2000 recognizes digital signatures in India and enables for its use. It has defined digital signature and provides for the use of digital signature to authenticate an electronic record (electronic record means data, record or data generated, image or sound stored, received or sent in an electronic form).

The IT Act further lays down that there shall be Certifying Authority who shall grant digital signature certificates to applicants. Such Certifying Authorities even have the power to suspend the certificate so granted.

To support and promote e-commerce and supplement the provision of the IT Act, the Indian Evidence Act, 1972 ('Act') is also being sought to be amended. The definition of evidence is amended to include electronic records. The terms digital signature, digital signature certificate, secure digital signature have also been included in the Act.

The Act shall further provide that the opinion of the Certifying Authority which has issued the Digital Signature Certificate shall be relevant in a court while adducing evidence in respect of on electronic document.

The IT Act shall also lay importance to secure digital signatures. It states that if a digital signature other than a secure digital signature is alleged to have been affirmed by any subscriber, such a digital signature would need to be proved.

A secure digital signature under the Act is a digital signature affixed by following security procedure as determined by the government. Such a digital signature should, on verfication, be unique to the subscriber affixing it, be capable of identifying such subscriber and be created in a manner under the exclusive control of the subscriber and linked to the electronic record in such a manner that if the electronic record was altered, the digital signature would be invalidated.

Thus, for the purpose of evidence in a court of law in India, a 'secure digital signature' would be accepted without its validity being challenged. With legal aid being given to digital signature by the IT Act and the proposed amendments to the Act, the use of digital signature to authenticate electronic transactions and contracts would soon be the norm. In the buzz of e-commerce today, another killer 'app' on the Internet-e-mails shall also use digital signatures to authenticate communication.

Thus, digital signatures are properly implemented, they will help us in a number of ways like preserving message integrity, minimizing the risk of undetected message tampering and forgery or false claims that a message was altered after it was sent; minimizing the risk of dealing with forgers or impostors or persons who escape responsibility by saying that they have been impersonated; and lastly, by retaining a

high degree of security, information can be sent over open, insecure and inexpensive networks (like the Internet where your e-mail can actually be handled by other parties before it reaches the intended recipient).

Cyber crime is increasing by leaps and bound with the increase of internet connectivity. Cyber crimes are crime related to information technology, electronic commerce etc. Cyber frauds, phreaks, hacking, obscene materials, viruses, harassment, privacy protection and intellectual property computer crimes committed via information highways on cyber space.

DOTCOMS, Get the legal thing right

The pot of gold at the end of a venture capital rainbow is still too strong a temptation. But a strange-dark phantom has now started spooking the dotcom opera. And, the apparition that casts a pall over the dotcom phenomenon is of legal nature. It's speciality it can strike any time, from any where. Content, contractual or service agreements, intellectual property rights (IPRs) all have to be made safe and secure from day one.

Most portals are too busy burning money-not their own, of course, provided by some munificent VC to bother about legal niceties. People have to be hired, context has to be source, the host server needs to be tied up, direly-needed eyeballs must be rustled up fast (since the second round of funding depends on that). Did you say legal due diligence? Get lost! Get me some more sticky eyeballs instead.

The portal (or, more likely, a website masquerading as a portal) is now up and running. Your friends, cousins, neighbours all troop in through the portal once in a while to see what you are up to. And then, suddenly, bang!! You get hit with a legal notice for something someone has put on the message board-offensive to 'public sensibilities' and requiring a painful trudge to the court. Or else, you have forgotten to register your trademark and you are now fighting a legal battle without any protective Armour.

So, whether you like it or not, legal eagles are indispensable lot for any net business, whether it is B2B or B2C. for those of you planning to set up new portals and without a clue about the legal jungle, here is a rough roadmap.

Legal Minefields for Dotcoms

Potential e-customers can also become potential legal liabilities. For example, a bond issue may be invited subscription on the web. The application forms can be down loaded and payments made off-line. "If non-residents wish to apply, the onus of complying with the exchange control regulations must be on them. Such terms and conditions must be prominently displayed on the website", mentions Sridhar Gorthi, partner with law firm Singh & Gorthi.

Content Control

Content is the adhesive which brings sticky customers. Loads of content, and different kinds of content, bringing long term customers. But even this is fraught with problems. The more the information which is accessible via the site, the greater the number of content providers, the more complex is the risk management process. The reason: content can accidentally defame a person or some of it may even be illegal in a few countries.

Pornography is prohibited under applicable criminal law. The Indian Penal Code (IPC) can hold liable even those persons who put obscene work into streams of commerce". "In fact, when a website containing objectionable content, is accessed from India, courts may hold that the owner of the site has committed an offence under the IPC. A grey cloud also hangs over adult chat sites".

Web Certification

The Information Technology Act, 2000 has thrown up some unexpected business avenues. Like Certification of identity which promises to be a money spinner for first movers. The Act could see big players like ICICI and HDFC in the fray for being appointed as certification agencies. These certificates would be for establishing identity and creditability of the person or entity transacting business on the web. These certificates would eventually be available for mobile phones as well. In order to bring forth key corner stones of trust worthy transactions into the e-commerce realm-integrity, authenticity, confidentiality and non-repudiation, the IT provides for a National level controller, which would be empowered to appoint certification Agencies which would issue certificates to individuals, corporate and Business entities. Trends indicate that Banks and consultancy firms, Accounting firms assure the role of certifying agencies. However, players like ICICI might be front rammers in the money game. In the US, such certificates can be obtained for $20. In India it may around Rs. 25000 for client certificate for transaction business on the web. Corporate and Business entities might pay between Rs. 15000 to 20000 for such a certificate (Reference-Hindustan Times, 14th June 2000).

Technology, like love, changes everything. With the ability to digitize any form of information, boundaries between the various forms of surveillance are disappearing with the application of information technology linking surveillance techniques into a near seamless web of surveillance. Developments in data processing suggest that the distinction between information and physical privacy is becoming more and more flimsy. The benefits of information technology are, of course, immurne and many facets of modern life are totally dependent upon the technology. While dangers can never be eliminated, however, proper regulation is in the interests of the technophile as much as the technophone. We live in uncertain, challenging and interesting times.